

PROJECT ABSTRACT [Form – 1] SESSION (2026-27)

PROJECT ID: SKIT/AI/2023-2027/07 **BRANCH:** Computer Science & Engineering **SECTION:** _____

TITLE OF PROJECT: Identifying URL-Based Attacks using IP Data

SDG MAPPING: SDG 9 – Industry, Innovation and Infrastructure

PROJECT TRACK: (Tick the appropriate one)

☒ R&D ☐ INNOVATION	☐ CONSULTANCY PROJECTS	☐ DEPARTMENT AS CLIENT PROJECTS	☐ CLIENT PROJECT	☐ HARDWARE-BASED PROJECTS
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MANDATORY EXTERNAL EVALUATION OF PROJECT (Tick the appropriate one)

☒ RESEARCH PAPER PUBLICATION	☐ PATENT FILING / GRANT	☐ POSTER COMPETITION
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PROBLEM STATEMENT:

Traditional blacklist-based security systems fail to detect newly generated (zero-day) malicious URLs and provide limited integration of URL, IP and threat-intelligence signals, along with a lack of secure, tamper-proof logging. This project proposes a hybrid cybersecurity framework that detects and classifies malicious URLs in real time by combining URL feature analysis, IP intelligence, domain reputation and live threat intelligence. Machine Learning drives intelligent decision-making, while Blockchain maintains immutable, tamper-evident audit logs — enabling proactive attack prevention, reliable threat monitoring and trustworthy digital-forensic analysis.

TECHNOLOGY STACK:

NAME OF TOOL / TECHNOLOGY	FRONTEND /BACKEND	SOFTWARE / HARDWARE	PURPOSE OF USE
React.js, Tailwind CSS, Chart.js, Axios, React Router	Frontend	Software	Responsive dashboard, URL-scanner UI, analytics charts and threat-report views
Python, FastAPI, Pydantic, Uvicorn	Backend	Software	REST APIs, request handling, URL validation and module orchestration
Scikit-Learn, XGBoost, Pandas, NumPy, WHOIS/DNS, Scapy, VirusTotal, AbuseIPDB	Backend	Software	Feature extraction, ML classification and threat-intelligence / IP-reputation validation
PostgreSQL, Redis, Solidity, Ganache, Web3.py, Docker	Backend / Database	Software	Data storage, caching, blockchain audit logging and CI/CD deployment

PROPOSED PROJECT SPRINTS (add more row if required):

S.No.	SPRINT NAME	PROPOSED USER STORY OF RESPECTIVE SPRINT
1	URL Intelligence Pipeline & Feature Engineering Framework	As a security analyst, I want the system to automatically extract lexical, host-based, domain and content features from any submitted URL, so that raw links are converted into a structured, ML-ready feature set for accurate threat evaluation.
2	Security Dashboard Architecture & Frontend Component Framework	As an end user, I want a responsive dashboard with a URL-scanner interface and reusable UI components, so that I can submit URLs and navigate results, history and reports through a clean, intuitive interface.
3	Threat Analysis Backend & API Orchestration Layer	As an API consumer, I want a FastAPI orchestration layer that validates URLs, coordinates the feature-extraction and detection modules and returns a unified analysis response, so that a single REST endpoint delivers consistent, low-latency threat verdicts.
4	Threat Classification Engine & Model Optimization Framework	As a security analyst, I want a trained and tuned ML model that classifies each URL as Benign, Phishing, Malware or Suspicious with a confidence-based risk score, so that malicious links are prioritized accurately with minimal false positives.
5	Security Analytics Dashboard & Visualization Engine	As a decision maker, I want interactive charts, threat-trend analytics and exportable reports (CSV / JSON / PDF), so that I can monitor detection metrics and generate audit-ready summaries.
6	Threat Intelligence Integration, Database Optimization & Deployment	As a system administrator, I want live VirusTotal / AbuseIPDB integration, Redis caching over an optimized PostgreSQL schema and blockchain-anchored audit logs deployed through a CI/CD pipeline, so that the platform runs as a tamper-proof, production-grade real-time detection service.

TEAM MEMBER DETAILS:

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Kartik Bhargava	8829924234	Backend, Databases, REST APIs, System Integration	Team Lead – Backend & Integration
Ishanvi Agarwal	6377373796	Machine Learning, Feature Engineering, Model Training	ML Engineer

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Diya Garg	_____	Frontend Development, Dashboard, Data Visualization	Frontend & UI/UX Developer

Verified & Approved By:

Mentor’s Name & Signature

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form – 2]

PROJECT ID: SKIT/AI/2023-2027/07 **BRANCH:** Computer Science & Engineering **SECTION:** ____

TITLE OF PROJECT: Identifying URL-Based Attacks using IP Data

TEAM LEAD Kartik Bhargava **SPRINT(s) NAME:** Sprint 3 & Sprint 6

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
As a developer, I want a FastAPI project skeleton with routing and config, so that all backend modules share a consistent foundation.	10-08-2026	05-09-2026	Initialized the FastAPI app, project structure, environment configuration, CORS and base routing.
As an API consumer, I want URL validation and request/response schemas, so that only well-formed URLs enter the pipeline.	06-09-2026	30-09-2026	Built Pydantic request/response models and a URL sanitization & validation layer with structured error handling.
As a system, I want an orchestration service, so that feature-extraction, threat-intel and ML modules run as one coordinated flow.	01-10-2026	25-10-2026	Implemented the orchestration service coordinating feature-extraction, threat-intelligence and ML detection modules.
As a frontend consumer, I want a single /analyze endpoint, so that one request returns a complete threat verdict.	26-10-2026	20-11-2026	Exposed the unified /analyze REST endpoint, wrote integration tests and wired the backend to the frontend.
As a security analyst, I want VirusTotal & AbuseIPDB integration, so that verdicts are backed by external threat intelligence.	21-11-2026	15-12-2026	Integrated VirusTotal and AbuseIPDB APIs with response normalization, retries and rate-limit handling.
As a user, I want fast repeat lookups, so that previously scanned URLs return instantly.	16-12-2026	15-01-2027	Added a Redis caching layer for repeated URL/IP lookups to cut latency and external API calls.
As an administrator, I want an optimized database, so that analysis data is stored and queried efficiently.	16-01-2027	10-02-2027	Normalized and indexed the Users, URL_Analysis, Threat_Intelligence and Audit_Logs tables in PostgreSQL.
As an auditor, I want tamper-proof logs and automated deployment, so that the system is trustworthy and production-ready.	11-02-2027	10-03-2027	Built the Solidity/Web3.py contract for immutable audit logging; containerized with Docker and set up GitHub Actions CI/CD.

MEMBER1 Ishanvi Agarwal **SPRINT(s) NAME:** Sprint 1 & Sprint 4

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
As a data engineer, I want benign/malicious URL datasets ingested, so that the model has quality training data.	10-08-2026	05-09-2026	Assembled benign and malicious URL datasets and built ingestion scripts.
As an analyst, I want lexical and host-based features, so that URL structure signals are captured.	06-09-2026	30-09-2026	Implemented extraction of lexical, host-based and length-based URL features.
As an analyst, I want domain and content features with WHOIS/DNS enrichment, so that reputation signals are included.	01-10-2026	25-10-2026	Added domain-age, registrar, DNS and content features through WHOIS/DNS enrichment.
As a data scientist, I want an ML-ready feature schema, so that models train on clean, consistent inputs.	26-10-2026	20-11-2026	Delivered a clean, versioned, ML-ready feature set with a preprocessing pipeline.
As a data scientist, I want baseline models, so that I have a performance benchmark.	21-11-2026	15-12-2026	Trained baseline Scikit-Learn classifiers and set up train/test splits and evaluation metrics.
As a security analyst, I want 4-class classification, so that URLs are labelled Benign/Phishing/Malware/Suspicious.	16-12-2026	15-01-2027	Built an XGBoost model for 4-class URL classification.
As a data scientist, I want tuned, validated models, so that accuracy is high and false positives are low.	16-01-2027	10-02-2027	Performed hyperparameter tuning and k-fold cross-validation to improve accuracy and reduce false positives.
As a backend, I want a packaged model with risk scoring, so that it can be served in production.	11-02-2027	10-03-2027	Implemented confidence-based risk scoring and exported the final model for backend serving.

MEMBER2 Diya Garg **SPRINT(s) NAME:** Sprint 2 & Sprint 5

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
As a developer, I want a React+Tailwind base with a component library, so that the UI is consistent and reusable.	10-08-2026	05-09-2026	Set up the React + Tailwind project and a reusable UI component library.
As an end user, I want a landing page and URL-scanner, so that I can submit URLs easily.	06-09-2026	30-09-2026	Built the landing page and URL-scanner input with validation UI.
As an end user, I want results, history and reports pages, so that I can review past scans.	01-10-2026	25-10-2026	Developed the results view, scan-history and reports pages.

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
As a web user, I want a responsive layout with live data, so that the dashboard adapts to any screen size.	26-10-2026	20-11-2026	Implemented responsive web layouts and API integration via Axios with React Router.
As a decision maker, I want analytics charts, so that I can see risk trends at a glance.	21-11-2026	15-12-2026	Integrated Chart.js for risk-score trend and detection-statistics charts.
As an analyst, I want threat-category and trend visualizations, so that I can spot patterns.	16-12-2026	15-01-2027	Built threat-category distribution and threat-trend visualizations.
As a user, I want exportable reports, so that I can share findings offline.	16-01-2027	10-02-2027	Implemented CSV/JSON/PDF report export from the dashboard.
As a presenter, I want a polished UI, so that the project demos cleanly.	11-02-2027	10-03-2027	Finalized UI/UX polish and prepared the dashboard for the project demonstration.

MEMBER3 — **SPRINT(s) NAME:** *Not applicable (three-member team)*

Verified & Approved By:

Mentor's Name & Signature